Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. *Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an* adherent point *of A if every open neighbourhood U of x satisfies*

$$U \cap A \neq \varnothing.$$

In the special case where X is a metric space with metric d, this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \varnothing.$$

Definition 2. *Let X be a topological space and $A \subseteq X$. The set A is called* dense *in X if every nonempty open set $U \subseteq X$ satisfies*

$$U \cap A \neq \varnothing.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. *Let X be a topological space and $A \subseteq X$. The set A is called* nowhere dense *in X if*

$$\operatorname{int}(\overline{A}) = \varnothing.$$

Baire Category Theorem

Theorem 1 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X. Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.*

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \qquad r_1 < 1.$$

Inductively, having chosen y_n, r_n, use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \qquad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \to 0$. Thus (y_n) is Cauchy, so by completeness $y_n \to y \in X$.

For each fixed n, all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \varnothing,$$

so the intersection is dense. $\square$

Remark 1. *From the definitions it follows that a set $B \subseteq X$ satisfies* $\operatorname{int}(B) = \varnothing \iff$ *$X \setminus B$ is dense in X. In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X.*

Corollary 1. *Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X. Then*

$$\operatorname{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \varnothing.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. $\square$